Hidden Polynomials

Questions 1–14

Question 1. Solve the following equations, giving your answer in exact form:

a) $5x^4 = 3 - 2x^2$
b) $3x - 16\sqrt{x} = 35$
c) $2x^{\frac{2}{3}} + 7x^{\frac{1}{3}} = 15$
d) $\frac{6}{x^2} + \frac{7}{x} = 5$
e) $7^x + 7^{2x} = 2$
f) $4^x - 3 \times 2^{x-1} + \frac{1}{2} = 0$
g) $3^{3x+1} + 20 \times 3^{2x} - 7 \times 3^x = 0$
h) $2^{x+1} + 11 = \frac{6}{2^x}$

Question 2. By first factorising, express $5^8 - 3^8$ as a product of its prime factors.

Question 3. Find the solutions to

$$\frac{2x^4+2}{5} = x^2$$

Question 4. Given

$$\mathrm{f}(x) = 8x^3 + \frac{1}{x^3}$$

solve $\mathrm{f}(x) = -9$

Question 5. Given

$$\mathrm{f}(x) = 2x^6 + \frac{1}{x^3} \text{ and } \mathrm{g}(x) = x^3 + 2$$

solve $\mathrm{f}(x) = \mathrm{g}(x)$

Question 6. Find the solutions to

$$\left(x - \frac{4}{x}\right)^2 + 8 = 6\left(x - \frac{4}{x}\right)$$

Question 7. Find the solutions to

$$\frac{5x^2}{\sqrt{x}} + 22 + \frac{24}{\sqrt{x}} = 12x + \frac{39x}{\sqrt{x}}$$

Question 8. Find the solutions to

$$3^{2x+1} + 20 \times 3^x + \frac{75}{3^{2x}} = 22 + \frac{220}{3^x}$$

Question 9.

Prerequisite knowledge for Q9–Q12: Exponentials and Logarithms

Solve the following equations, giving your answer in exact form:

a) $3 \times 2^{2x} + 7 \times 2^x = 6$

b) $2e^{3x} + 3e^{2x} - 17e^x + 12 = 0$

c) $4 \times 3^x + 13 = \frac{2}{3^x} + \frac{15}{3^{2x}}$

Question 10.

a) Solve

$$\frac{4}{(x-2)^2} - \frac{7}{x-2} = 2$$

b) Hence, solve

$$\frac{4}{(5^x-2)^2} - \frac{7}{5^x-2} = 2$$

giving your answer in exact form.

Question 11. Solve the following equations, giving your answer in exact form:

a) $\log_2(x+3) = 2\log_2 x - \log_2 4$

b) $\log_4(5x+2) = \log_4(13x-6) - 2\log_4 x$

c)

$$-1 + \log_5(x^4+4)$$
$$= \log_5 x + \log_5(3x+14-5x^2) - \log_5 6$$

Question 12. Solve for the following equation giving your answer in exact form

$$\ln x + \frac{4}{\ln x} = \frac{\ln(x^4)}{\ln x}$$

Question 13.

Prerequisite knowledge for Q13–Q14: Solving Trigonometric Equations

Solve

$$\sin^2 x = \frac{7 - 13\sin x}{2}$$

for $0 \leq x \leq 360°$

Question 14. Solve

$$\frac{\cos^2 x(3\cos x + 11)}{20} = \cos x$$

for $0 \leq x \leq 360°$